

# William Rodriguez

# Two-House Buyer Recommendations

*Data-Driven Home Recommendations Based on Market Hypotheses*

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## Persona

# Buyer William Rodriguez

### What he wants

William Rodriguez is house-hunting for two people and wants to buy two homes: a non-renovated country house he can purchase at the right time, and a fast-moving, centrally located city house.

## Workflow

# Data Analysis Pipeline



## Our Hypotheses

- H1: The more renovated, the higher the price
- H2: The farther from downtown, the bigger the lot
- H3: The more floors, the smaller the lot

# H1: Price vs Renovation Status

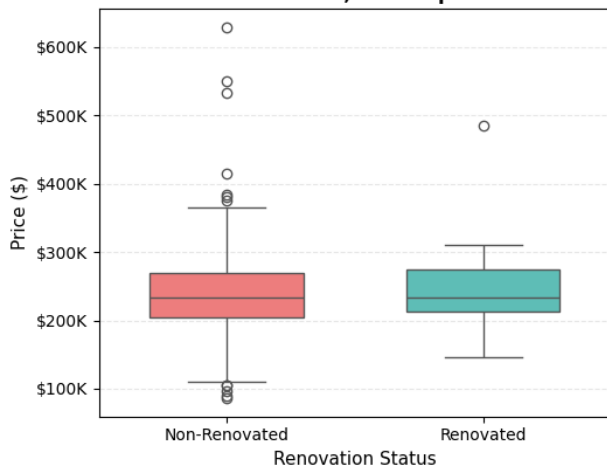
## Country House Analysis

*The more renovated, the higher the price*

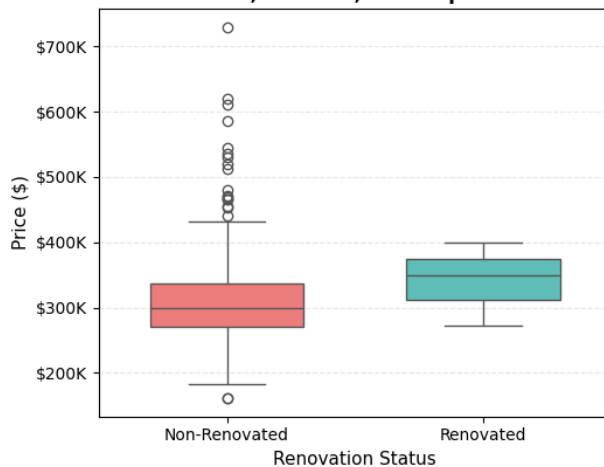
# Price by Renovation Status

Renovation Effect by Property Size in Rural Areas

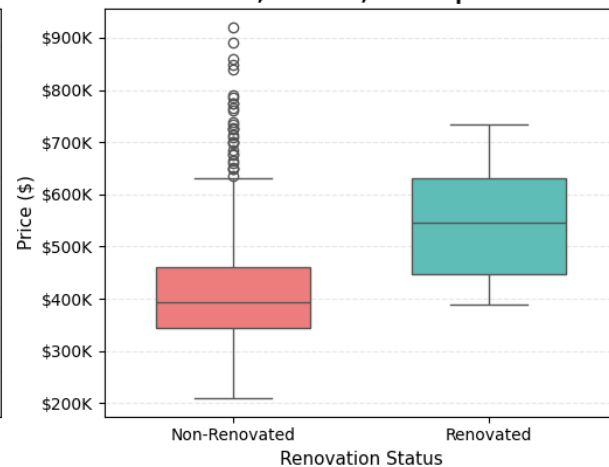
Small Properties  
650.0 - 1,670.0 sqft



Medium Properties  
1,677.0 - 2,260.0 sqft



Large Properties  
2,270.0 - 5,820.0 sqft



Key Findings: The more renovated, the higher the price

# House Selection Criteria

### Selection Criteria

| Criteria    | Target          |
|-------------|-----------------|
| Location    | ✓ Rural         |
| Renovation  | ✓ Non-Renovated |
| Size Range  | ✓ Medium        |
| Zipcode     | ✓ 98022         |
| Homes Found | ✓ 2             |

# Two Rural Homes for William

### House #1 - Zip 98022

|             |                   |
|-------------|-------------------|
| Price       | <b>\$195,000</b>  |
| Living Area | <b>1,580 sqft</b> |
| Bed / Bath  | <b>3 / 1.75</b>   |
| Year Built  | <b>1979</b>       |
| Lot Size    | <b>7,875 sqft</b> |

#### Why William will love it:

- ✓ Rural, quiet, no close neighbors
- ✓ Non-renovated = lower price
- ✓ Tied for lowest price in candidate pool

### House #2 - Zip 98022

|             |                   |
|-------------|-------------------|
| Price       | <b>\$195,000</b>  |
| Living Area | <b>1,570 sqft</b> |
| Bed / Bath  | <b>3 / 1.75</b>   |
| Year Built  | <b>1991</b>       |
| Lot Size    | <b>8,459 sqft</b> |

#### Why William will love it:

- ✓ Newer build, slightly larger lot
- ✓ Same price as House #1
- ✓ Tied for lowest price in candidate pool

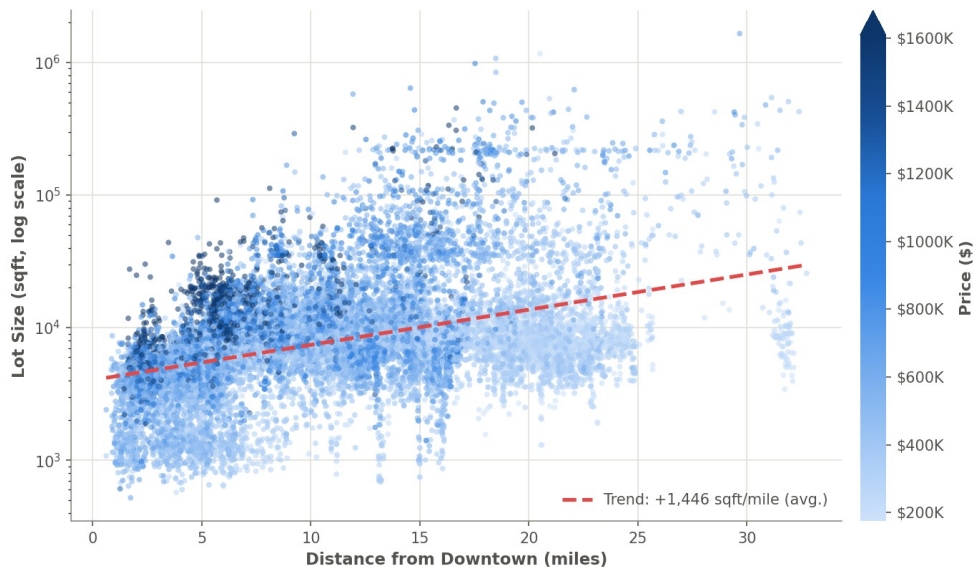
## H2: Lot Size vs Distance From Downtown Seattle

*The farther from downtown, the bigger the lot*

# Lot Size Grows With Distance (Colored by Price)

## H2: Lot Size Grows With Distance From Downtown Seattle

Color = sale price · n = 20,896 houses



### Key Findings:

- The bigger the distance from downtown, the bigger the lot size
- Trend: +1,446 sqft/mile (avg.)

# House Selection Criteria

### Selection Criteria

| Criteria       | Target           |
|----------------|------------------|
| Suburban Dist. | ✓ ~14.6 mi       |
| Rural Dist.    | ✓ ~29.7 mi       |
| Suburban Lot   | ✓ 641,203 sqft   |
| Rural Lot      | ✓ 1,651,359 sqft |
| Homes Found    | ✓ 2              |

# Suburban & Rural Options

### Suburban - Zip 98070

|             |                     |
|-------------|---------------------|
| Price       | <b>\$849,900</b>    |
| Distance    | <b>14.56 mi</b>     |
| Lot Size    | <b>641,203 sqft</b> |
| Living Area | <b>2,280 sqft</b>   |
| Bed / Bath  | <b>2 / 2</b>        |

#### Why William will love it:

- ✓ Quiet and peaceful, not too far
- ✓ Large lot for outdoor space
- ✓ Confirmed by H2: closer-in large lot

### Rural - Zip 98022

|             |                       |
|-------------|-----------------------|
| Price       | <b>\$700,000</b>      |
| Distance    | <b>29.66 mi</b>       |
| Lot Size    | <b>1,651,359 sqft</b> |
| Living Area | <b>1,300 sqft</b>     |
| Bed / Bath  | <b>4 / 1</b>          |

#### Why William will love it:

- ✓ Maximum quiet and solitude
- ✓ Huge lot: maximum privacy
- ✓ Cheaper than the suburban option

# H3: Floors vs Lot Size

## Urban City House Analysis

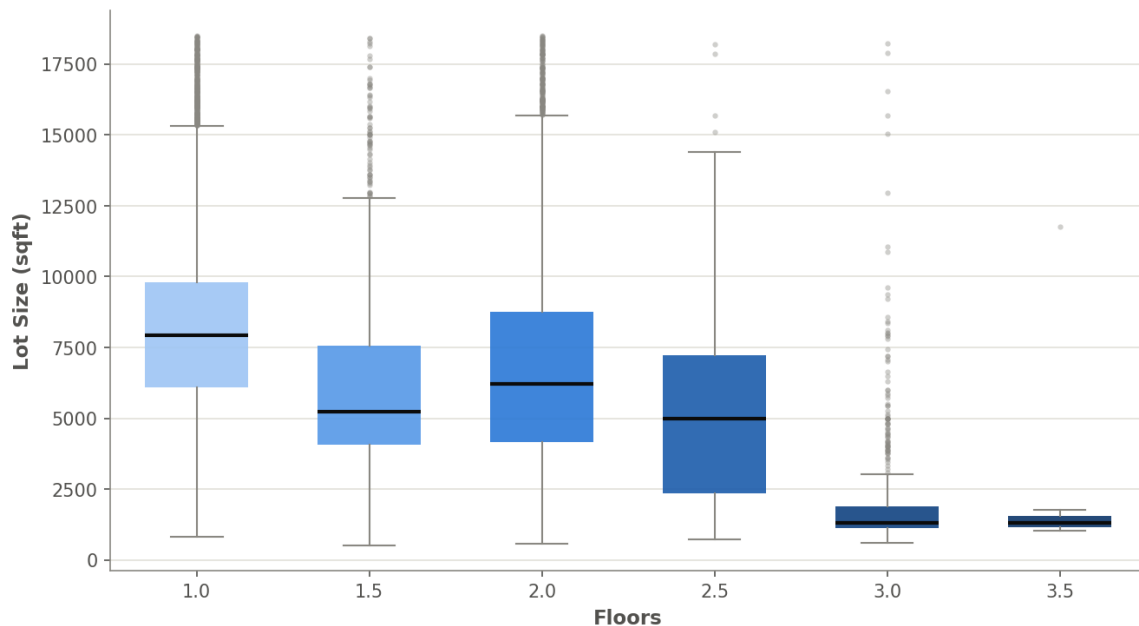
*The more floors, the smaller the lot*

## H3 Chart

# Lot Size by Floor Count (Urban Houses)

### H3: More Floors Goes With Smaller Lots (Urban Houses)

Spearman  $r = -0.316$  ·  $p < 0.0001$  ·  $n = 18,404$  houses



#### Key Findings:

- The more floors, the smaller the lot
- Pearson's  $r = -0.316$

# Floors vs Lot Size - Selection Criteria

## Selection Criteria

|             |                                              |
|-------------|----------------------------------------------|
| Floors      | ✓ $\geq 2$                                   |
| Lot Size    | ✓ $\leq 7,200$ sqft (median)                 |
| Location    | ✓ Urban (RUCA 1)                             |
| Sort Order  | ✓ Price descending                           |
| Assumption  | ✓ Pricier sells faster<br>(proxy for demand) |
| Homes Found | ✓ 2 (tied at \$2.95M)                        |



18,404 Urban houses analyzed



4,706 match criteria; top 2 by price shown

## H3 Recommendations

# Two Compact City Homes

### House #1 - Zip 98107

|             |             |
|-------------|-------------|
| Price       | \$2,950,000 |
| Floors      | 3.0         |
| Lot Size    | 5,722 sqft  |
| Living Area | 4,340 sqft  |
| Bed / Bath  | 5 / 4.5     |

#### Why William will love it:

- ✓ Genuinely vertical, compact footprint
- ✓ Well below the Urban median lot
- ✓ Tied for highest price - assumed fastest sale

### House #2 - Zip 98199

|             |             |
|-------------|-------------|
| Price       | \$2,950,000 |
| Floors      | 2.0         |
| Lot Size    | 5,884 sqft  |
| Living Area | 4,470 sqft  |
| Bed / Bath  | 4 / 4.25    |

#### Why William will love it:

- ✓ Slightly larger living area
- ✓ Well below the Urban median lot
- ✓ Tied for highest price - assumed fastest sale

# Q&A

*Thank you for listening and paying attention*